**Laxmi Charitable Trust's**
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**
Department of Information Technology (B.Sc.I.T Semester V)
APPLIED BUSINESS
# Practical 7

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| Class: TYIT | Batch: 1 |
| Date of Assignment: 03-10-26 | Date/Time of Submission: 03-10-26 |

Practical 7: Sampling and Sampling Distributions
Part A: Survey and Sampling
1. Select a real-life business topic for conducting a survey.
2. Prepare a questionnaire containing at least 5 questions related to the selected topic.
3. Collect responses from at least 50 people.
4. Store the collected data in a CSV file or Pandas DataFrame.
5. Display the first five records of the collected data.
6. Identify the population and sample for the survey.
7. Identify the sampling technique used to select the sample.
8. Identify the numerical and categorical variables in the collected data.
9. Select one numerical variable for further analysis.
10. Calculate the sample size, mean, median, and standard deviation of the selected variable.
11. Create a suitable chart to display the collected data.
12. Write two observations from the survey data.

**CODE:**

```
# Practical 7: Sampling and Sampling Distributions
# Part A: Survey and Sampling

# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

# 1. Select a real-life business topic
# Topic: Online Shopping Customer Satisfaction Survey

# 2. Questionnaire
# Q1. What is your age?
# Q2. What is your gender?
# Q3. How often do you shop online?
# Q4. How satisfied are you with online shopping? (1-5)
# Q5. How much do you usually spend on online shopping per month?

# 3. Responses have been collected from 60 people.

# 4. Load the collected data from CSV file
df = pd.read_csv("Online_Shopping_Survey.csv")
```

```python
# 5. Display the first five records
print("First Five Records:")
print(df.head())

# 6. Identify Population and Sample

population = "All people who shop online"
sample = "60 people who participated in the survey"

print("\nPopulation:", population)
print("Sample:", sample)

# 7. Identify the sampling technique

sampling_technique = "Convenience Sampling"

print("Sampling Technique:", sampling_technique)

# 8. Identify numerical and categorical variables

numerical_variables = ["Age", "Satisfaction_Rating", "Amount_Spent"]
categorical_variables = ["Gender", "Shopping_Frequency"]

print("\nNumerical Variables:")
print(numerical_variables)

print("\nCategorical Variables:")
print(categorical_variables)

# 9. Select one numerical variable for further analysis

selected_variable = df["Amount_Spent"]

print("\nSelected Numerical Variable: Amount_Spent")

# 10. Calculate sample size, mean, median and standard deviation

sample_size = selected_variable.count()
mean_value = selected_variable.mean()
median_value = selected_variable.median()
std_value = selected_variable.std()

print("\nSample Size:", sample_size)
print("Mean Amount Spent:", mean_value)
print("Median Amount Spent:", median_value)
print("Standard Deviation:", std_value)
```

```python
# 11. Create a suitable chart

plt.figure(figsize=(8, 5))
plt.hist(df["Amount_Spent"], bins=8, edgecolor="black")
plt.title("Distribution of Monthly Online Shopping Amount")
plt.xlabel("Amount Spent")
plt.ylabel("Number of Respondents")
plt.show()

# 12. Observations

print("\nObservations:")
print("1. The amount spent by respondents varies across different ranges.")
print("2. Most respondents spend a moderate amount on online shopping each month.")
```

**OUTPUT**

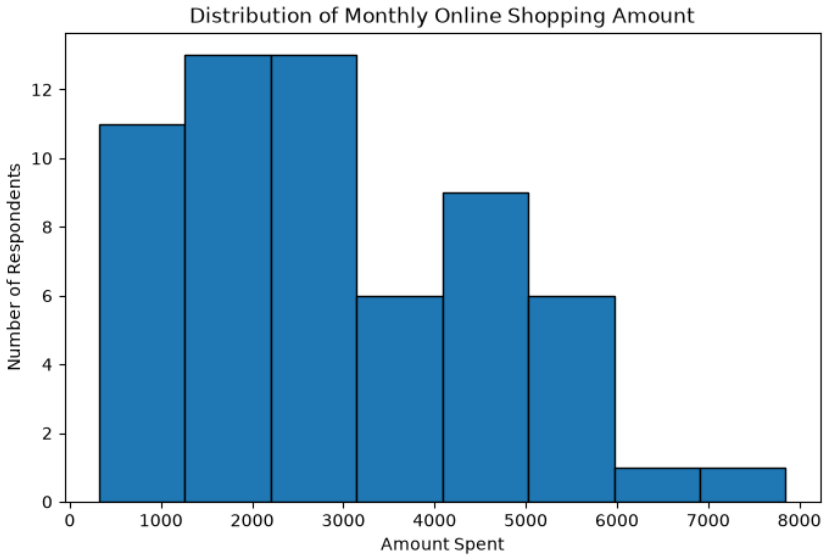
```
First Five Records:
   Respondent_ID  Age  Gender Shopping_Frequency  Satisfaction_Rating  \
0              1   21  Female            Monthly                    3
1              2   21  Female            Monthly                    2
2              3   24  Female            Monthly                    2
3              4   35    Male             Weekly                    4
4              5   23    Male             Weekly                    3

   Amount_Spent
0          1271
1          1083
2          1514
3          1853
4          4557

Population: All people who shop online
Sample: 60 people who participated in the survey
Sampling Technique: Convenience Sampling

Numerical Variables:
['Age', 'Satisfaction_Rating', 'Amount_Spent']

Categorical Variables:
['Gender', 'Shopping_Frequency']
...
Sample Size: 60
Mean Amount Spent: 2952.616666666667
Median Amount Spent: 2785.0
Standard Deviation: 1715.6496925404451
```




```
Observations:
1. The amount spent by respondents varies across different ranges.
2. Most respondents spend a moderate amount on online shopping each month.
```

Part B: Sampling Distribution and Central Limit Theorem

13. Select a numerical variable from the collected data.
14. Select a sample of 30 observations randomly from the data.
15. Calculate the mean of the selected sample.
16. Repeat the random sampling process 100 times.
17. Calculate and store the mean of each sample.
18. Display the first 10 sample means.
19. Calculate the mean of all the sample means.
20. Calculate the standard deviation of the sample means.
21. Create a histogram of the sample means.
22. Observe the shape of the sampling distribution.
23. Compare the original sample mean with the mean of the sampling distribution.
24. Compare the sampling distributions obtained for different sample sizes.

25. Calculate the standard error of the sample mean.
26. Explain how the Central Limit Theorem is demonstrated through the experiment.
27. Use the sample mean to estimate the population mean.
28. Write a conclusion based on the sampling distribution and Central Limit Theorem.

**CODE**

```
# Practical 7: Sampling and Sampling Distributions
# Part B: Sampling Distribution and Central Limit Theorem

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv("Online_Shopping_Survey.csv")

# 13. Select a numerical variable
data = df["Amount_Spent"]

print("Selected Numerical Variable: Amount_Spent")

# Original population/sample mean
original_mean = data.mean()

print("Original Mean:", original_mean)


# 14. Select a sample of 30 observations randomly
sample_30 = data.sample(n=30, random_state=42)

print("\nRandom Sample of 30 Observations:")
print(sample_30.values)


# 15. Calculate the mean of the selected sample
```

```
sample_mean = sample_30.mean()

print("\nMean of Selected Sample of 30:", sample_mean)


# 16 & 17. Repeat random sampling 100 times
# and calculate/store the mean of each sample

sample_means = []

for i in range(100):
    sample = data.sample(n=30)
    mean = sample.mean()
    sample_means.append(mean)

print("\nTotal Sample Means:", len(sample_means))


# 18. Display the first 10 sample means

print("\nFirst 10 Sample Means:")

for i in range(10):
    print(i + 1, ":", sample_means[i])


# 19. Calculate the mean of all sample means

mean_of_sample_means = np.mean(sample_means)

print("\nMean of All Sample Means:", mean_of_sample_means)


# 20. Calculate the standard deviation of the sample means

std_sample_means = np.std(sample_means, ddof=1)

print("Standard Deviation of Sample Means:", std_sample_means)


# 21. Create a histogram of the sample means

plt.figure(figsize=(8, 5))

plt.hist(sample_means, bins=10, edgecolor="black")

plt.title("Sampling Distribution of Sample Means")
```

```
plt.xlabel("Sample Mean of Amount Spent")
plt.ylabel("Frequency")

plt.show()


# 22. Observe the shape of the sampling distribution

print("\nObservation of Sampling Distribution:")
print("The sampling distribution is approximately bell-shaped and centered around the
mean.")


# 23. Compare original sample mean with
# the mean of the sampling distribution

print("\nComparison:")
print("Original Mean:", original_mean)
print("Mean of Sampling Distribution:", mean_of_sample_means)

difference = abs(original_mean - mean_of_sample_means)

print("Difference:", difference)


# 24. Compare sampling distributions for different sample sizes

sample_sizes = [10, 20, 30, 40]
distribution_means = {}

for size in sample_sizes:

    means = []

    for i in range(100):
        sample = data.sample(n=size)
        means.append(sample.mean())

    distribution_means[size] = means


# Create separate histograms for different sample sizes

for size in sample_sizes:

    plt.figure(figsize=(8, 5))
```

```
    plt.hist(
        distribution_means[size],
        bins=10,
        edgecolor="black"
    )

    plt.title(
        "Sampling Distribution - Sample Size " + str(size)
    )

    plt.xlabel("Sample Mean")
    plt.ylabel("Frequency")

    plt.show()


# 25. Calculate the Standard Error of the Sample Mean

population_std = data.std(ddof=1)
sample_size = 30

standard_error = population_std / np.sqrt(sample_size)

print("\n25. Standard Error of Sample Mean:")
print("Standard Error:", standard_error)


# 26. Explain Central Limit Theorem

print("\n26. Central Limit Theorem:")
print("The Central Limit Theorem states that when repeated random")
print("samples are taken from a population, the distribution of")
print("sample means becomes approximately normal as the sample")
print("size increases.")


# 27. Use sample mean to estimate population mean

estimated_population_mean = mean_of_sample_means

print("\n27. Estimated Population Mean:")
print("Estimated Population Mean:", estimated_population_mean)


# 28. Conclusion

print("\n28. Conclusion:")
```

```
print("The sampling distribution of Amount_Spent was created by")
print("taking 100 random samples of 30 observations.")
print("The sample means were centered around the original mean.")
print("As the sample size increased, the sampling distribution")
print("became more concentrated around the population mean.")
print("This demonstrates the Central Limit Theorem.")
```

**OUTPUT**

```
Selected Numerical Variable: Amount_Spent
Original Mean: 2952.616666666667

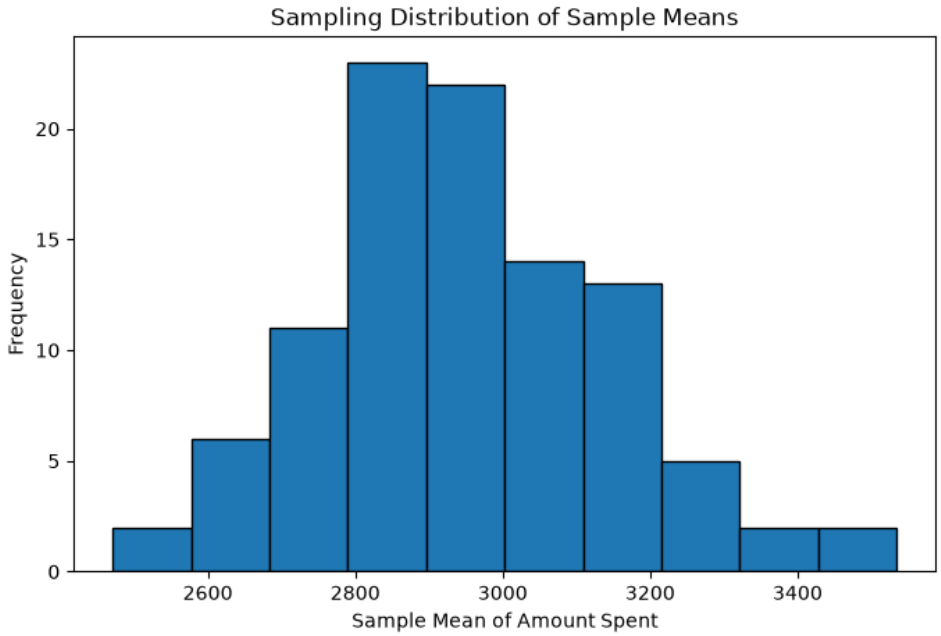
Random Sample of 30 Observations:
[1271 4617 1479 5786  769 5672 3284  680 1440 3417 2299 3090 4859 1853
 4257 2933 5870 2250 5256 4557 3551  960 6206 3073 2872 4278 5558 4145
 1228 1794]

Mean of Selected Sample of 30: 3310.133333333333

Total Sample Means: 100

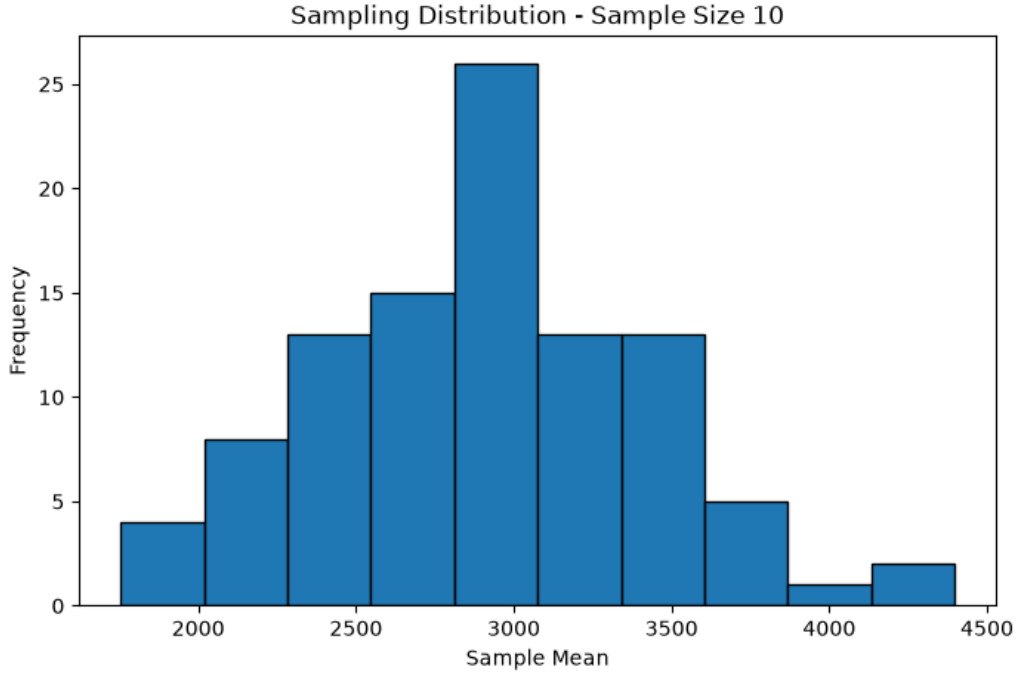
First 10 Sample Means:
1 : 2954.9666666666667
2 : 3396.0
3 : 2767.766666666667
4 : 2887.633333333333
5 : 2620.2
6 : 2644.7
7 : 3186.0666666666666
8 : 2805.4
9 : 2866.733333333333
10 : 2834.8333333333335

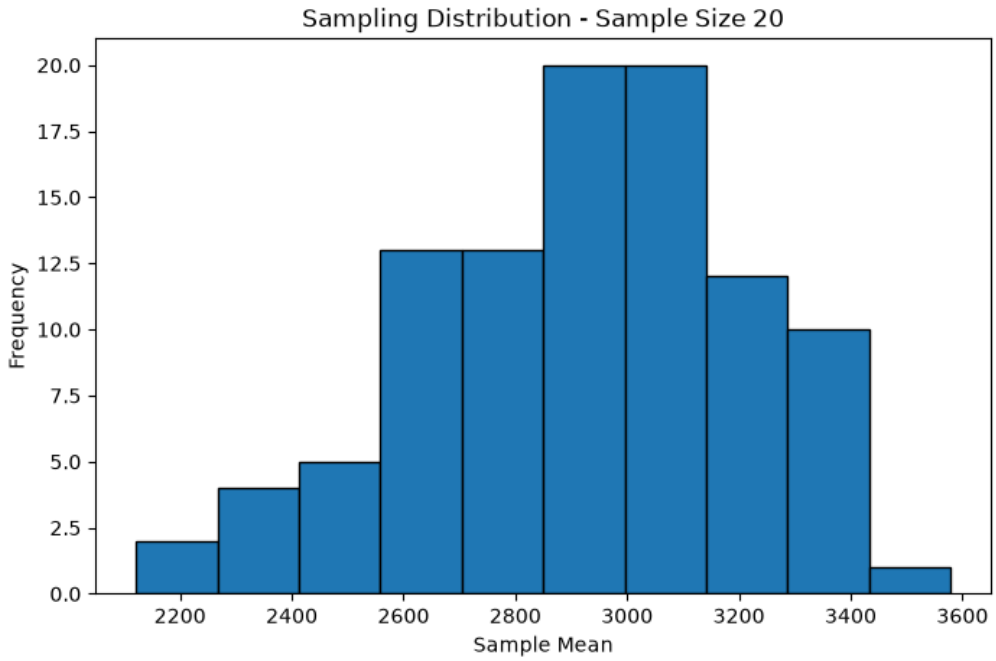
Mean of All Sample Means: 2949.549666666667
Standard Deviation of Sample Means: 199.59955902187863
```

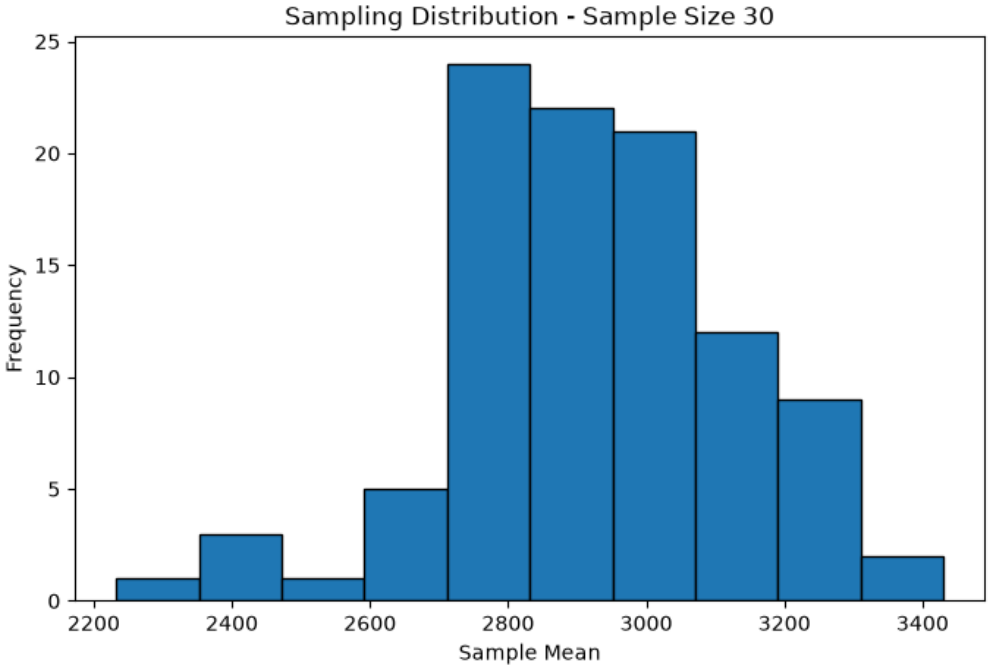


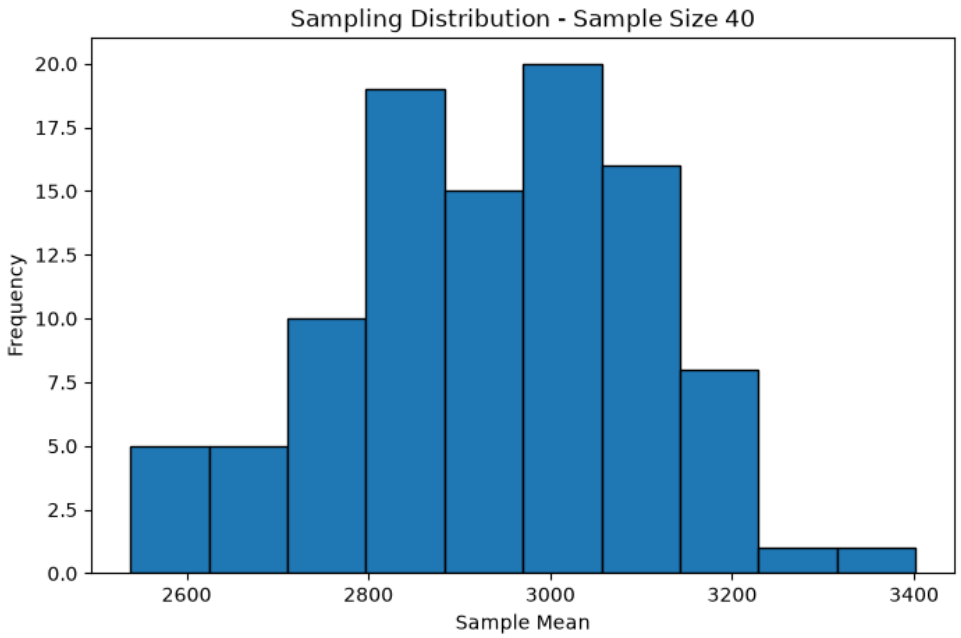

```
Observation of Sampling Distribution:
The sampling distribution is approximately bell-shaped and centered around the mean.

Comparison:
Original Mean: 2952.616666666667
Mean of Sampling Distribution: 2949.549666666667
Difference: 3.0670000000000073
```




Sampling Distribution - Sample Size 20
Frequency
Sample Mean
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
2200
2400
2600
2800
3000
3200
3400
3600


Sampling Distribution - Sample Size 30
Frequency
Sample Mean
0
5
10
15
20
25
2200
2400
2600
2800
3000
3200
3400




```
25. Standard Error of Sample Mean:
Standard Error: 313.2333457937348

26. Central Limit Theorem:
The Central Limit Theorem states that when repeated random
samples are taken from a population, the distribution of
sample means becomes approximately normal as the sample
size increases.

27. Estimated Population Mean:
Estimated Population Mean: 2949.549666666667

28. Conclusion:
The sampling distribution of Amount_Spent was created by
taking 100 random samples of 30 observations.
The sample means were centered around the original mean.
As the sample size increased, the sampling distribution
became more concentrated around the population mean.
This demonstrates the Central Limit Theorem.
```